PLC Practical Examinations: Ladder Logic Programs

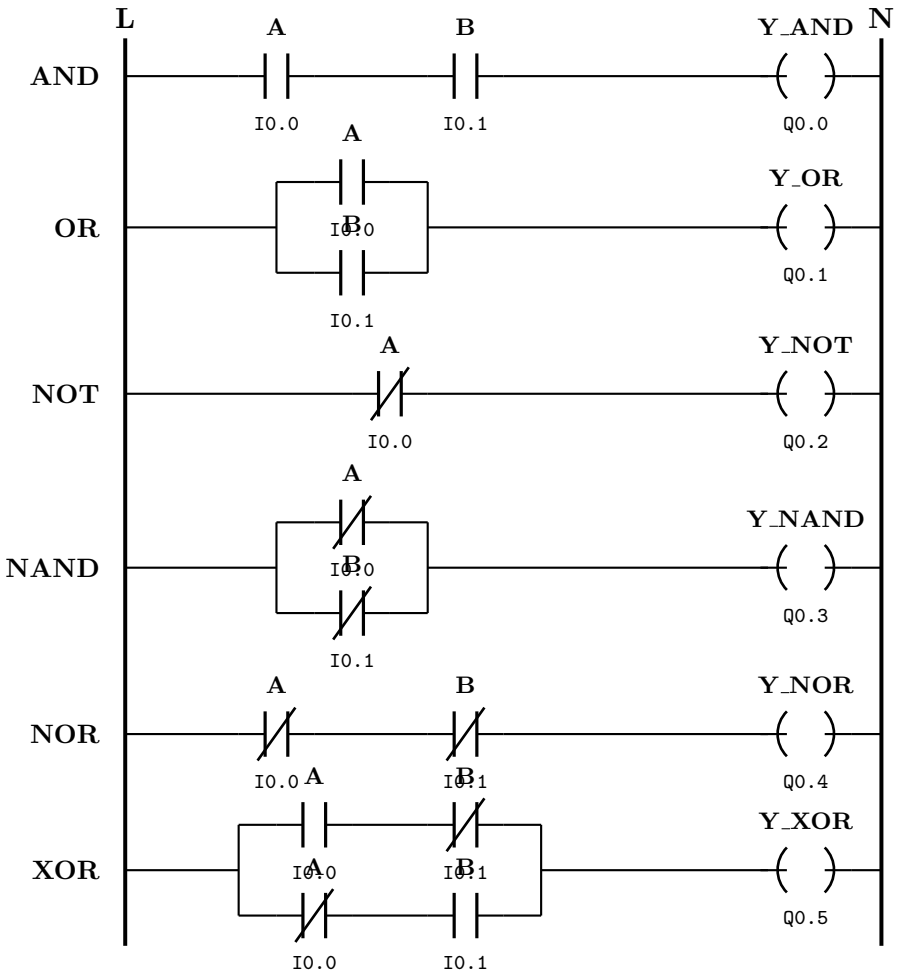
Department of Electronics and Telecommunication

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Practical 1: Implementation of Basic Logic Gates

Aim: To design and implement basic logic gates (AND, OR, NOT, NAND, NOR, XOR, XNOR) using ladder diagrams.

1.1 AND, OR, NOT, NAND, NOR Gates



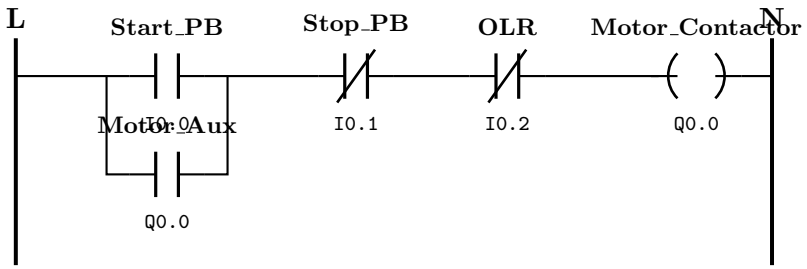
Practical 2: Development of Ladder Program for DOL Starter

Aim: Start/stop control of a 3-phase induction motor with seal-in (latching) contact and thermal overload protection.

Address Assignment

Device Name	PLC Address	Type
Start Push Button	I0.0	NO Contact
Stop Push Button	I0.1	NC Contact
Overload Relay (OLR)	I0.2	NC Contact
Contactor Coil (Motor)	Q0.0	Output Coil

Ladder Diagram



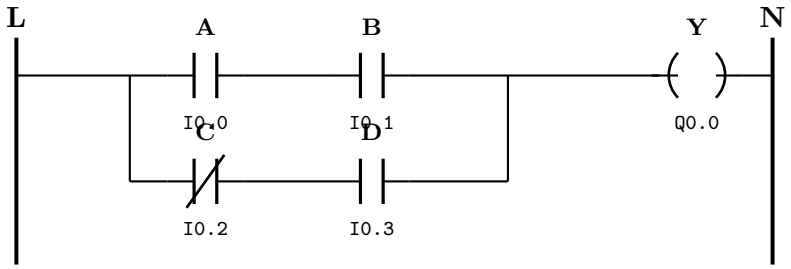
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Practical 3: Implementation of Boolean Expression

Aim: To implement and verify the Boolean function:

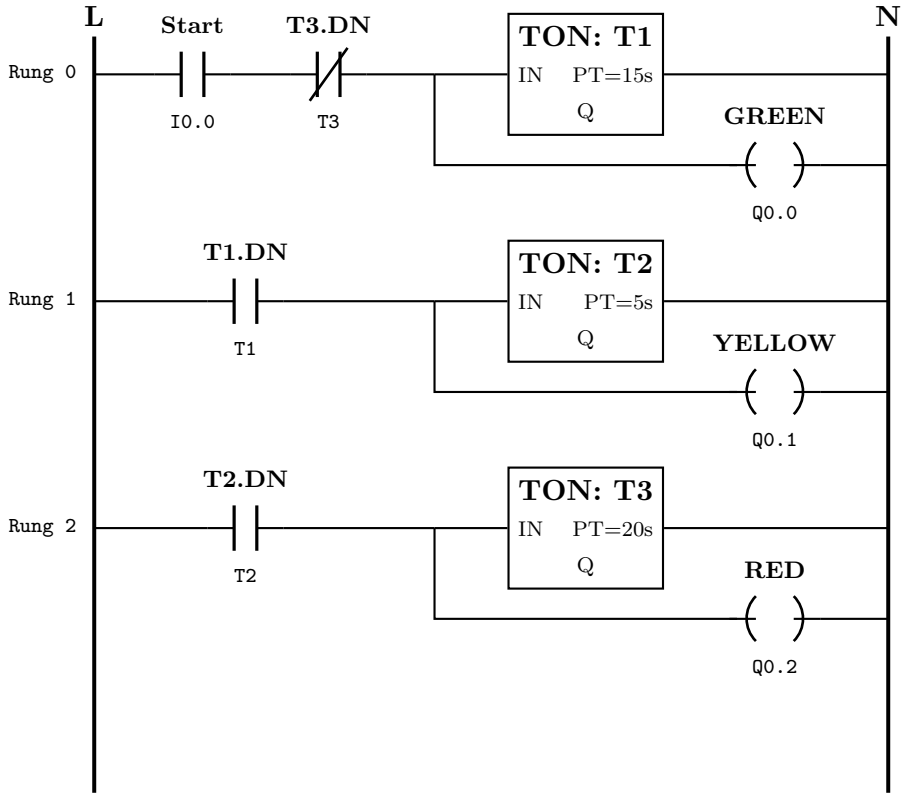
$$Y = (A \cdot B) + (\overline{C} \cdot D)$$

Ladder Diagram



Practical 4: Traffic Light Control Using Timers (TON)

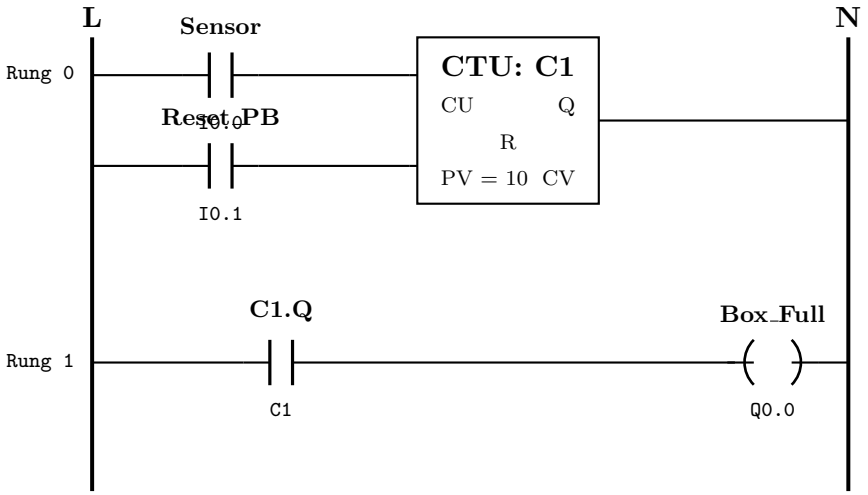
Aim: Automatic sequential switching of Green (15 s), Yellow (5 s), and Red (20 s) lights using On-Delay Timers.



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Practical 5: Object Counter (Bottle Filling Plant)

Aim: Increment a counter with each bottle detected by a proximity sensor; stop the conveyor / illuminate carton-full indicator once preset is reached.



Practical 6: Pneumatic Cylinder Control Using PLC

Aim: Automatic forward extension and retraction of a double-acting cylinder using 5/2-way solenoid valves and magnetic limit switches.

Address Assignment

Tag Name	Address	Description
Start_PB	I0.0	Cycle Start Push Button
LS_Home	I0.1	Cylinder Retracted Position (Limit Switch 1)
LS_Extended	I0.2	Cylinder Extended Position (Limit Switch 2)
SOL_Extend	Q0.0	Solenoid Valve A (Extend Cylinder)
SOL_Retract	Q0.1	Solenoid Valve B (Retract Cylinder)

Ladder Diagram

